STYLMORPH

Major Project

Guide:

Prof. Jincy Showri

By Group 13:

Devi Nandana
Neeraj S
Sanjay Kumar P M
Vinayakrishnan U

Outline

Overview

The proposed platform allows users to create realistic 3D avatars of themselves by entering body measurements like height, weight, and body shape.

Users can virtually try on clothes from various e-commerce platforms using product links. The clothing items are mapped accurately onto the 3D model.

The system also uses AI to recommend matching outfits and color combinations based on the user's profile. Available as both web and mobile application for wide accessibility.

Problem Statement

Lack of personalized and realistic try-on systems:

- Most e-commerce platforms use 2D overlays or generic 3D avatars that do not reflect unique body measurements, shape, or skin tone.
- Shoppers are unable to visualize how clothing will truly look or fit on their own bodies.
- Leads to confusion and dissatisfaction when garments appear different in reality.

Problem Statement (contd.)

Increased return rates and reduced purchase confidence:

- Inaccurate size predictions and unrealistic previews $\rightarrow$ products don't meet expectations.
- High volume of returns increases operational costs.
- Reduced user trust in the platform.
- Missed sales opportunities and poor overall shopping experience.

Existing System

- Uses static 2D profiles (age,gender,preferences).
- Relies on browsing purchase history for recommendations.
- Product visualization is limited to 2D images/videos.
- Provides only basic personalization (similar items, trending products)

Proposed System

- 3D model for each user based on their own body measurements.
- Using an e-commerce product link, users can check wearable items.
- Unique models for each user.
- Introducing color tone–based 3D models.
- Ease of finding dress combinations.

Relevance

- Reduces product returns by helping users choose the right fit and size.
- Enhances shopping experience with personalized, realistic virtual try-ons.
- Matches clothes to skin tone for better style and visual appeal.
- Saves time with smart outfit and color combination suggestions.

References

- Virtual fitting using 3D avatar and garment simulation: https://doi.org/10.1186/s40691-020-00214-z
- AI-powered smart mirror: A virtual dressing room using deep learning and AR. IEEE ICCCI 2021: https://doi.org/10.1109/ICCCI50826.2021.9402465

Thank You!